
Model Predictive Control Practicals

Model Predictive Control : stabilisation and tracking

Evaluation : you will be asked to provide an electronic report (pdf format). In this report, you answer the questions, you add screen shots to illustrate and comment your results. Also, you will be asked to upload a compressed folder containing your Matlab files. These files must be executable, well structured and your code must be clear and commented. You have until 3rd of January 2022 to render your work on Moodle.

1 Stabilization

Consider the system defined by the transfer function :

$$\frac{Y(s)}{U(s)} = \frac{1}{s(s+1)}$$

1. Considering a Zero-Order-Hold, find a discrete-time state space representation of system for a sampling period $T_s = 1s$:

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{u}(t)\end{aligned}\tag{1}$$

where $t \in \{0, 1, 2, \dots\}$ is the discrete time.

2. Considering the initial condition $\mathbf{x}(0) = [8 \ -5]^T$. We want to program an MPC controller that permit to stabilize the system with a horizon $N = 10$. To do so, we consider to minimize the cost function

$$\sum_{k=0}^{N-1} \left(\mathbf{x}_k^T \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}_k + \frac{1}{2} u_k^2 \right) + \mathbf{x}_N^T \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}_N$$

under the constraints $-4 \leq u_k \leq 4$ for $k = \{0, \dots, N-1\}$. Complete the Matlab files *MPC_Stabilization_Init.m* and *MPC_Stabilization.slx* in order to implement the MPC controller.

2 Tracking

Considering the same system, we want to track a step reference signal r_k .

1. Formulate the tracking problem using the input increment formulation. The aim is to minimize the cost function

$$\sum_{k=0}^{N-1} \left(2(\mathbf{y}_{k+1} - r_0)^T (\mathbf{y}_{k+1} - r_0) \right) + \frac{1}{2} \Delta \mathbf{u}_k^T \Delta \mathbf{u}_k$$

under the constraints $-2 \leq \mathbf{u}_k \leq 2$ and $-1 \leq \Delta \mathbf{u}_k \leq 1$ for $k = \{0, \dots, N-1\}$.

Note that r_0 is the value of the reference signal in the beginning of the horizon. We consider again $\mathbf{x}(0) = [8 \ -5]^T$ and $N = 10$.

2. Complete the Matlab files *MPC_Tracking_Init.m* and *MPC_Tracking.slx* in order to implement the MPC tracking controller.

3 Integral Action (bonus question)

1. Are the previous two solutions robust with respect to an input disturbance ?
2. Find a MPC controller which permits to solve the tracking problem while rejecting the input perturbation